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Low-temperature electronic specific heat of $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$. Evidence for a d wave superconductor

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Abstract

The electronic specific heat C_{el} was studied at $T \leq 10$ K both on $La_{2-x}Sr_xCuO_4$ samples and on $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples below and above y_c at which the superconductivity disappears. In superconducting $La_{2-x}Sr_xCuO_4$ samples, the C_{el} contains only a T^2 component at $T \ll T_c$ in a x range between 0.14 and 0.22, suggesting a gapless superconductor with line nodes. A T linear component of C_{el} appears in superconducting $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ ($y < y_c$) samples, and its coefficient, γ, increases markedly with Zn content. In samples with various Zn contents, the relation between $T_c/T_c(y=0)$ and $\gamma/\gamma(y \geq y_c)$ can be understood in terms of strong impurity scattering in a d wave superconductor.

1. Introduction

Since the discovery of high-T_c cuprates, numerous studies on their superconducting properties have been carried out. In high-T_c cuprates, partial substitution of non-magnetic Zn for Cu leads to a marked suppression of T_c, as do magnetic Ni, Co, Mn and Fe [1–4]. It has been reported for Zn doped cuprates that local magnetic moments are induced on Cu sites around Zn atoms [3–7]. It has therefore been claimed that the marked suppression of T_c in Zn doped samples can be ascribed to magnetic pair breaking due to the local moments [4], as in the conventional s wave superconductors. However, it was reported recently that magnetic pair breaking due to the local moments induced by Zn doping is not strong enough to account for the marked suppression of T_c, and suggested that the superconductivity will be of d wave pairing type [8–10] and that the suppression of T_c is likely due to ordinary potential scattering [9,10]. When impurities are doped into a superconductor with d symmetry, the electronic spectral weight at the Fermi level E_F will increase markedly from zero even though its doping level is very small. It is therefore interesting to investigate the spectral weight at E_F in superconducting cuprates whose Cu atom is partly substituted for with small amounts of other elements.

The $La_{2-x}Sr_xCuO_4$ system is potentially one of the most appropriate systems for the study of impurity doping at the Cu site because there is only a single kind of Cu site. In $La_{2-x}Sr_xCuO_4$ samples of high quality, the T linear term of the electronic specific heat C_{el}, whose coefficient is proportional to the electronic spectral weight (density of states) at E_F, van-

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ishes at $T \ll T_c$ [11]. For the study of the symmetry of the superconducting order parameter, it is therefore of interest to clarify whether the T linear term of C_{el} is revived at $T \ll T_c$ when a small amount of nonmagnetic impurity is doped at the Cu site of $La_{2-x}Sr_xCuO_4$. Furthermore, the T dependence of the C_{el} at $T \ll T_c$ will also give important information on the symmetry of the superconducting order parameter. In a d wave superconductor the C_{el} at $T \ll T_c$ exhibits a power-law T dependence while in a s wave superconductor one has an exponential dependence, as is well known.

In the present study, we studied the electronic specific heat in detail on both $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples. Preliminary results have been reported elsewhere [12].

2. Experimental

Samples of $La_{2-x}Sr_xCuO_4$ were prepared using a solid-state reaction of stoichiometric amounts of 5N La_2O_3, 5N $SrCO_3$ and 5N CuO. The reaction and sintering were performed at 1030°C–1070°C for 48 h with an intermediate grinding, followed by oxygen annealing at 600°C for 24 h and slow cooling to room temperature.

Samples of $La_{2-x}Sr_xCu_{1-y}Zn_yCuO_4$ were prepared in the same procedures as those for $La_{2-x}Sr_xCuO_4$. In order to examine whether substitution for Cu with Zn is successful or not, the tetragonal-to-orthorhombic transition temperature T_0 was investigated. This is because chemical substitution in $La_{2-x}Sr_xCuO_4$ samples is expected to change the transition temperature T_0 significantly [13]. The T_0 was determined here from the anomaly in the temperature derivative of the resistivity [14]. The T_0 thus determined increases in proportion to the Zn content up to 6%, the maximum Zn content examined. This result indicates that the partial substitution for Cu with Zn is successful in the present sample preparation.

The specific-heat capacity was measured by a conventional adiabatic heat pulse method within a T range from 2 K to 10 K. The quantity of the sample used for the measurement was usually about 3 g. The diamagnetic transition curve of the superconductivity was measured using a SQUID susceptometer.

3. Results and discussion

In Fig. 1, typical diamagnetic transition curves are shown for $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples. The superconducting transition temperature T_c is defined here as the temperature at which a straight line extrapolated from the steepest part of the diamagnetic transition curve crosses the zero level. The critical Zn content y_c where the superconductivity totally disappears depends on the Sr content; the y_c values of the present samples for $x=0.10$, 0.16, 0.18, 0.2 and 0.22 are 2%, 3.5%, 4%, 4% and 4%, respectively. The destruction of superconductivity at y_c was also confirmed in measurements on the resistivity. Fig. 2 shows the T dependence of the resistivity for $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ sample with $y \simeq y_c$. As seen in Fig. 2, no superconductivity appears in the samples for $y \simeq y_c$ except a filamentary superconductivity in a few samples. It is also worth noting in Fig. 2 that the resistivity for $x \gtrsim 0.16$ shows a metallic behavior down to the lowest temperature examined, while the resistivity for $x \lesssim 0.16$ exhibits localization behavior of the carriers at low temperatures, probably, on account of random potentials due to impurity doping. This result indicates that the localization of carriers plays an important role in the suppression of superconductivity in Zn doped samples for $x<0.16$, but

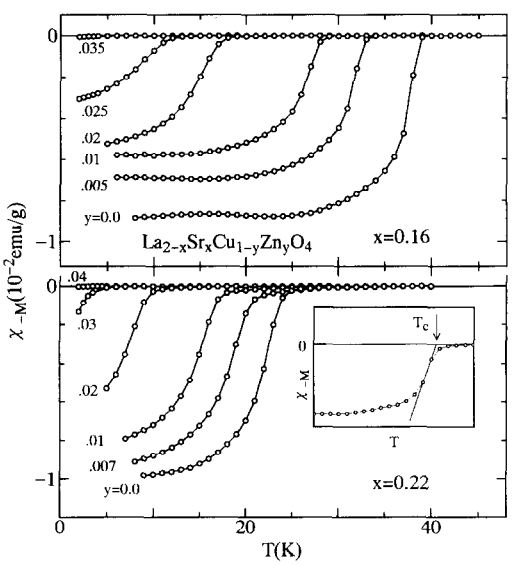


Fig. 1. Superconducting diamagnetic transition for $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ with $x=0.16$ and 0.22.

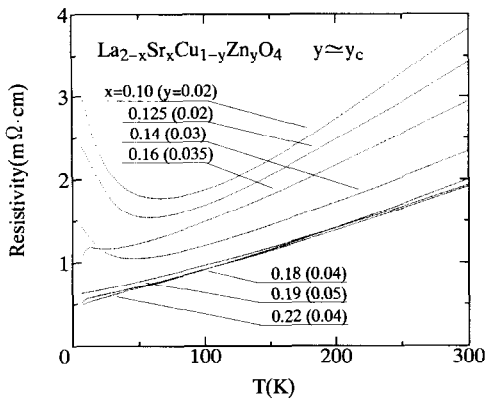


Fig. 2. Temperature dependence of the resistivity for $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ ($y \simeq y_c$).

does not crucially in Zn doped samples for $x \gtrsim 0.16$.

In Fig. 3, typical specific-heat results are shown for both $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ with a C/T versus T^2 plot. In samples for $La_{2-x}Sr_xCuO_4$ ($x=0.2$ and 0.22), the variation of C/T with T^2 follows a downward curve rather than a straight line at low temperatures. The C/T versus T^2 curves for these samples can be reproduced by the sum of both T^2 and T^3 components below ~ 8 K, as shown in Fig. 4(b). Results for $x=0.16$ and 0.18 also can be fitted to the sum of both components (Fig. 4(a)). However, the specific heat C for $x=0.16$ and 0.18 is dominated mainly by the T^3 component because the T^2 component is relatively small, and so the C/T versus T^2 curves resemble a straight line rather than a downward curve at low temperatures. On the other hand, data for non-superconducting $La_{2-x}Sr_xCu_{1-y}Zn_yCuO_4$ ($y \geq y_c$) samples can be fitted by using the following expression:

$$C = \gamma_0 T + \beta T^3 , \qquad (1)$$

below ~ 8 K, except at very low temperatures, as shown in Fig. 4. It should be noted in Fig. 4 that the coefficient of the T^3 term, β, is the same as that of the T^3 component contained in superconducting $La_{2-x}Sr_xCuO_4$ samples. The T^3 term is expected to be due to thermal excitation of phonons since the specific heat C_{ph} of the phonon system will show a T^3 dependence below a few percent of the Debye temperature θ_D, as observed in ordinary systems. The β value observed will give $\theta_D = 420$–450 K on the basis of the Debye model, which is in good agreement with

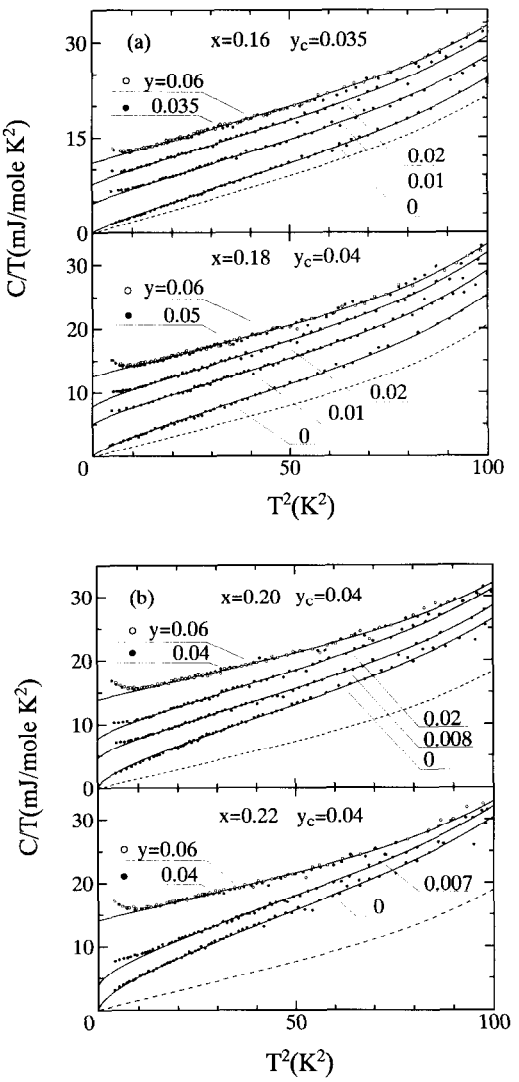


Fig. 3. Specific heat of $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ in a C/T vs. T^2 plot for (a) $x=0.16$, 0.18, (b) $x=0.20$, 0.22. The broken line represents the phonon term C_{ph}. The solid lines represent the results of fitting the data to Eq. (2) in the text for $y < y_c$ and to $C = \gamma_N T + C_{ph}$ for $y \geq y_c$.

those (430–480 K) determined from ultrasonic-velocity and neutron-diffraction experiments on the present system [15,16]. Furthermore, the temperature (8 K) below which the T^3 component appears in the present study corresponds to 1.8–1.9% of θ_D, as expected. On the other hand, the T^2 component contained in superconducting $La_{2-x}Sr_xCuO_4$ sam-

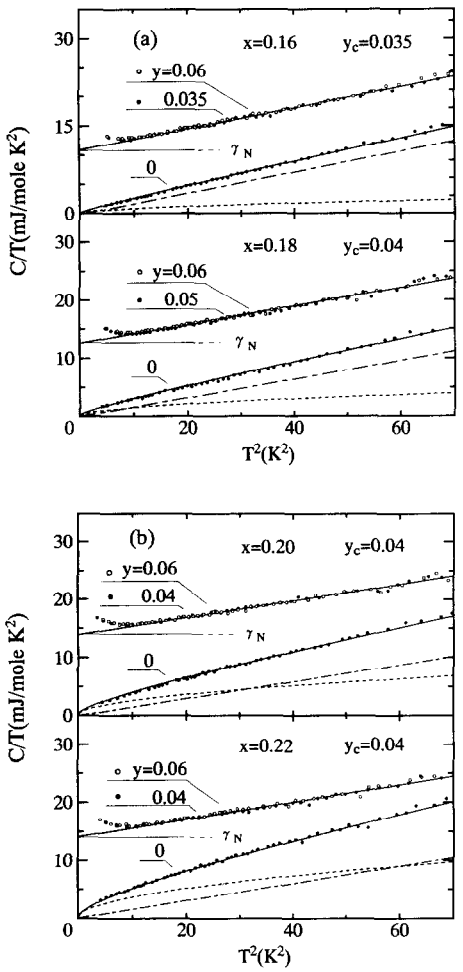


Fig. 4. Specific heat at $T \lesssim 8.5$ K for superconducting $La_{2-x}Sr_xCuO_4$ and non-superconducting $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ ($y > y_c$) with (a) $x=0.16$, 0.18, (b) $x=0.20$, 0.22. The solid lines for $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ ($y \geq y_c$) represent Eq. (1) in the text. The dotted line and the dash-dotted line show the αT^2 and the βT^3 term, respectively. The solid lines for $y=0$ represent the sum of the αT^2 and the βT^3 terms.

ples should be attributable to the electronic system. A superconducting electronic specific heat C_{el}^s of a power-law T dependence is familiar for exotic superconductivity of heavy-fermion systems such as $CeCu_2Si_2$, UPt_3, URu_2Si_2 and so on [17–19].

In non-superconducting $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ ($y \geq y_c$) samples, the C_{ph} can be obtained by subtracting the electronic T linear term, $\gamma_0 T$, from the observed C over the whole T range examined, except below ~ 3.5 K. As seen in Fig. 3, the specific heat of the Zn doped sample upturns below ~ 3.5 K on account of local magnetic moments induced by Zn doping [20–23]. Then, the C_{ph} at $T \lesssim 3.5$ K was estimated in this study by extrapolating the straight line which fits to the data at higher temperatures. The C_{ph} thus obtained up to 10 K is represented by dotted lines in Fig. 3. The C_{ph} of the present system remains unchanged when the superconducting state sets in, as mentioned above. Therefore, in superconducting $La_{2-x}Sr_xCuO_4$ samples, we can obtain the electronic specific heat C_{el}^s at $T < T_c$ by removing the C_{ph} from the observed C. In Fig. 5, the C_{el}^s thus obtained up to 10 K, the highest temperature examined, is shown for various Sr contents. It is noted in Fig. 5 that the C_{el}^s shows a T^2 dependence up to 10 K, and its coefficient, α, increases with increasing x in the x range above $x \simeq 0.15$.

It is clear from C/T versus T^2 plots for Zn doped samples (Fig. 3) that the C/T value extrapolated to $T=0$ increases markedly with increasing Zn content. This fact means that the electronic T linear component, whose coefficient is related directly to the density of states at E_F, $N(E_F)$, will become large with Zn content in a Zn doped superconducting sample. On the assumption that the electronic specific heat C_{el}^s of superconducting $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples can be reproduced by the sum of both a T linear and a T^2 term, fitting to the data was carried out using the following equation,

$$C(T) = \gamma T + \alpha T^2 + C_{ph}(T)\,, \qquad (2)$$

in order to separate the T linear component from the

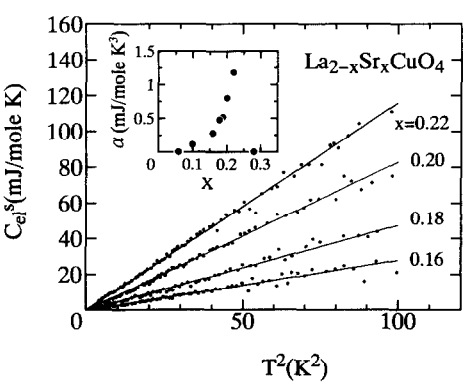


Fig. 5. Electronic specific heat C_{el}^s for superconducting $La_{2-x}Sr_xCuO_4$. The solid line represents a function αT^2. The coefficient, α, is shown as a function of x in the inset.

observed C. The phonon term C_{ph} was taken here to be the same as that observed for $y \geq y_c$ and to be independent of y. Experimental results on samples whose T_c's are much higher than 10 K can be fitted well to Eq. (2) over a wide temperature range up to 10 K, as shown in Fig. 3. The γ value obtained in the fitting is shown as a function of y in Fig. 6, as well as T_c. The γ value, namely $N(E_F)$, at $T \ll T_c$ increases markedly with increasing Zn content (Fig. 6) while it is zero for $La_{2-x}Sr_xCuO_4$ ($0.15 \leq x \leq 0.22$). The increase in γ becomes small above y_c. In particular, the γ value of Zn doped samples for $x \gtrsim 0.16$ becomes independent of y in the y range $y \geq y_c$ within the experimental scattering. So the γ value at $y \geq y_c$ can be taken as the value for the normal state, γ_N. On the other hand, in samples with $x < 0.16$, the γ_N was defined here by a linear extrapolation of γ values at $y \geq y_c$ back to $y = 0$, as indicated by the dotted line in Fig. 6. The γ_N value thus determined, together with T_c, is shown in Fig. 7 as a function of x. The present x dependence of γ_N is in qualitative agreement with that estimated from the specific heat above T_c by Loram et al., except a x range around $x = 0.115$ [24]. There exists a dip around $x = 0.115$ in the present γ_N versus x curve. This dip will be related to that observed around $x = 0.115$ in the T_c versus x curve.

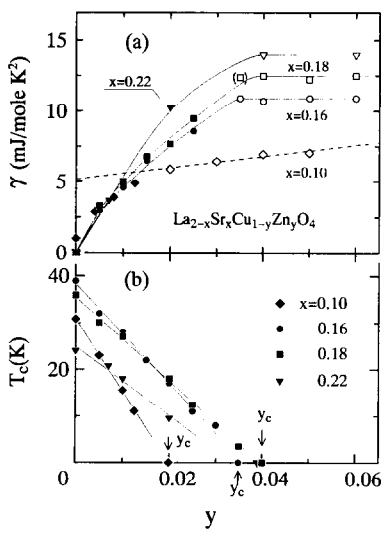


Fig. 6. (a) The γ value of $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples plotted as a function of y both below y_c (closed symbol) and above y_c (open symbol). The open-square symbol in the parentheses shows the γ value estimated from the data above T_c, although the Zn content of the sample is below y_c. The dotted line for $x = 0.1$ represents a linear extrapolation of γ to $y = 0$. (b) The T_c of $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ samples plotted as a function of y.

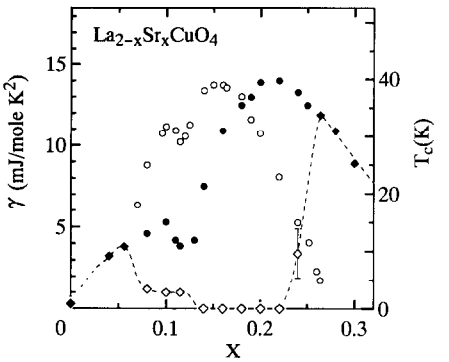


Fig. 7. Sr content dependence of T_c (open circles), γ value in the superconducting (open rhombuses) and normal state (closed circles), and γ value of non-superconducting samples for $La_{2-x}Sr_xCuO_4$ (closed rhombuses).

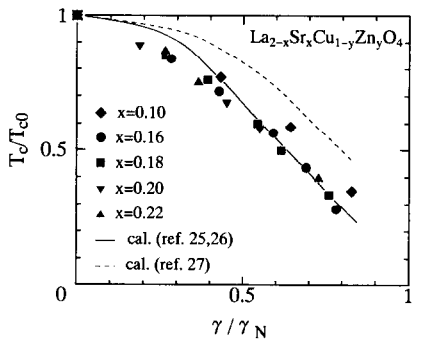


Fig. 8. $T_c/T_c(y=0)$ vs. γ/γ_N plots for $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$. The solid and broken lines represent the theoretical results for a d wave superconductor with line nodes [25–27].

The γ value at $T \ll T_c$ is markedly restored in Zn doped samples even if T_c decreases only a little, as seen in Fig. 6. Such a tendency has been known as a feature for a gapless superconductor doped with impurities whose scattering is in the unitarity limit. In Fig. 8, relations between $T_c/T_c(y=0)$ and γ/γ_N are shown for samples with x ranging from 0.1 to 0.22, together with the theoretical results for a gapless superconductor with line nodes calculated by two different groups [25–27]. A sample for $La_{2-x}Sr_xCuO_4$ ($x = 0.1$) exhibits a finite residual γ value at $T = 0$, and so the difference between both residual γ values for $y = 0$ and for $y \neq 0$ was regarded here as the intrin-

sic restoration of γ due to Zn doping. As seen in Fig. 8, the experimental results are in good agreement with the theoretical curve given by Miyake et al. [25,26], while in qualitative agreement with that calculated by Prohammer et al. [27]. The residual value of the density of states $N(E_F)$ at $T \ll T_c$ was studied from the Knight shift of NMR experiments by Ohsugi et al. on various superconducting cuprate systems [28,29], and it has been reported that the relation between the residual $N(E_F)$ and T_c also agrees with the theoretical result given by Miyake et al.

In a gapless superconductor with line nodes, the C_{el}^{s} will exhibit a T^2 dependence at low temperatures. In this case, the coefficient of the T^2 term, α, is given by

$$\alpha \propto \frac{N(E_F)}{\Delta_0} \propto \frac{\gamma_N}{\Delta_0}, \tag{3}$$

where Δ_0 is the amplitude of the superconducting gap. Here, by assuming that $La_{2-x}Sr_xCuO_4$ system is a gapless superconductor with line nodes, the x dependence of Δ_0 was estimated from the measured α and γ_N values (the inset of Fig. 5 and Fig. 7) by using Eq. (3). The x dependence of Δ_0 thus estimated is shown in Fig. 9, together with T_c. The Δ_0 will decrease monotonically with increasing Sr content above $x=0.15$, as well as the critical temperature T_c.

It should be stressed here that the Sr content at which the γ_N value reaches the maximum is different from the optimal Sr content for T_c, as seen in Fig. 7. If the coupling constant associated with the superconductivity is independent of x, the T_c will be proportional to $N(E_F)$, i.e., γ_N. Experimentally these results are not the case. This fact means that the coupling constant in $La_{2-x}Sr_xCuO_4$ system will significantly depend on the Sr content x, namely the hole concentration within the Cu–O plane.

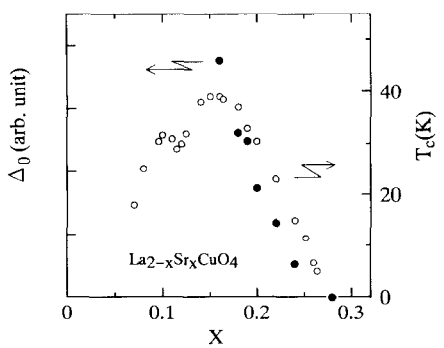


Fig. 9. Sr content dependence of T_c (open circle) and the superconducting gap Δ_0 (closed circle) estimated by using Eq. (3).

4. Summary

In superconducting $La_{2-x}Sr_xCuO_4$ samples, the electronic T linear term, γT, is absent at low temperatures below T_c in the x range between 0.15 and 0.22. Instead there exists an electronic T^2 component below T_c, suggesting a d wave superconductor with line nodes. In the $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ system, the γ value at $T \ll T_c$ increases markedly with increasing Zn content and T_c decreases largely. The relation between $T_c/T_c(y=0)$ and γ/γ_N can be understood in terms of strong impurity scattering in a d wave superconductor. Both results of the electronic specific heat on $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Sr_xCu_{1-y}Zn_yO_4$ indicate that the superconductivity of the $La_{2-x}Sr_xCuO_4$ system is of a d wave pairing type.

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